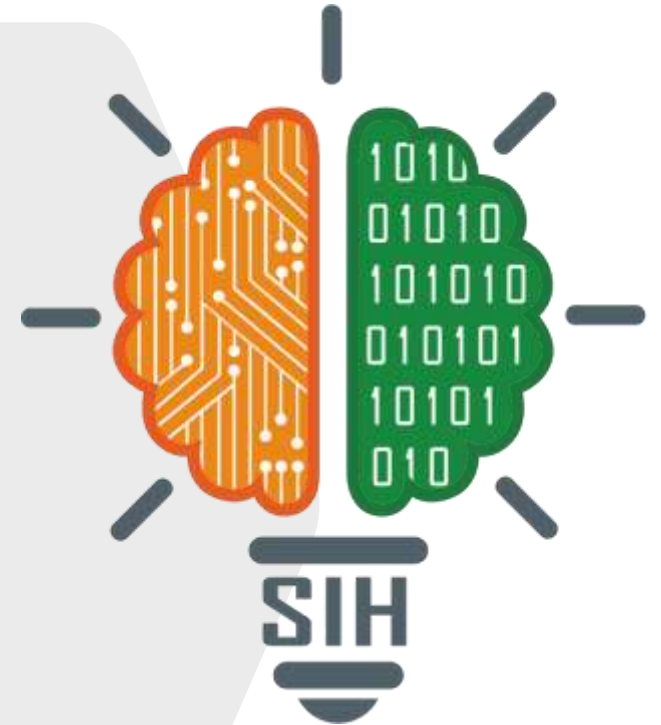


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26164**
- **Problem Statement Title – Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)**
- **Theme – Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID – 139892**
- **Team Name – MetaD Bandits**



WHY NOW

Adversaries are **harvesting encrypted data today** to unlock it when a quantum computer arrives. India's DST roadmap already sets **CII migration milestones for 2027-2029**, yet most teams cannot even see where RSA/ECC lives. Today's scanners hand back a **flat CBOM dump**. **ECDAT turns discovery into decisions**: what breaks first, what it costs to fix, and proof of what was checked.

Illustrative. Mosca.

$$X + Y > Z$$

X – Data-shelf time
Y – Migration Time
Z – Deadline

X – Data-shelf time

ARCHIVAL X~10 y
ballots, patient records**OPERATIONAL** X~5 y
business databases**SHORT-TERM** X~1.5 - 0 y
caches, session tokens

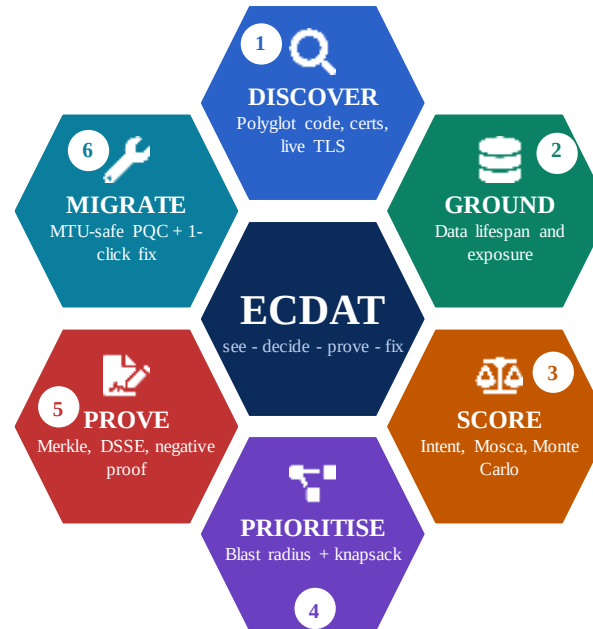
Y – Migration Time

5 years**4 years****2 years**

Z – Deadline

12 years
10+5=15>12 → **At Risk****10 years**
5+4=9<10 → **At Runtime****6 years**
1.5+2=3.5>6 → **At Runtime****Challenges & Problems**

- Invisible Crypto:** Classical Cryptography hides in code, configs, containers, certificates and libraries; no single inventory.
- Guessed Data Lifespan:** One blanket "10 years" for every asset makes Mosca's $X + Y > Z$ uncomputable.
- Primitive Flooding:** A cache-key SHA-256 is flagged like RSA guarding patient records; teams drown in alerts.
- Hidden Blast Radius:** One weak root CA or shared library exposes every downstream service; 5,000 findings, no fix order.
- Unprovable Reports:** Unsigned JSON can be edited, and "100% scanned" is a claim no vendor can prove.
- PQC Size Shock:** A 3,309 B ML-DSA-65 signature (51.7x ECDSA) overflows 64 B buffers and is dropped on real networks.

Proposed Solution: The ECDAT Loop

Numbers 1-6 match the problems on the left. Each stage removes one blind spot and hands a verified result to the next.

Innovation & Uniqueness**Schema-to-Lifespan**

Infers data lifespan X from SQL DDL, Prisma and ORM retention. No spreadsheet guessing.

*fixes problem 1***Intent-Aware Risk**

DSIS lattice gives cache and ETag hashes zero quantum weight; CAMS prices the refactor.

*fixes problem 2***Epidemic R0 Graph**

Finds superspreader CAs and libraries; one upgrade immunises every downstream service.

*fixes problem 3***Budget-Fit Sprint Planner**

Ranks fixes by risk removed per developer-week and fills a fixed sprint budget (Pareto frontier).

*fixes problem 4***Pre-Flight MTU Prober**

Predicts PQC packet drops (ML-DSA-65 needs 4 TCP segments) before code ships.

*fixes problem 5***Evidence-Graded Honesty**

E0-E5 evidence levels, an Unknowns Ledger and a signed, scoped negative certificate.

fixes problem 6

Methodology & Process

- 1 Perimeter & Network Probe:** declare the audit boundary, quarantine binaries and keystores, measure path MTU (1,500 B vs < 1,280 B).
- 2 Polyglot Discovery:** contract engines read JS/TS, Go, Rust, Java, Python, Ruby, manifests, Nginx/Apache configs, X.509 certs, live TLS and eBPF probes.
- 3 Context Grounding:** infer data lifespan X from SQL/ORM schemas and exposure P(HNDL) from Docker and Kubernetes manifests.
- 4 Semantic Scoring:** classify intent (DSIS), agility (CAMS 0-3) and buffer hazards; compute Mosca Y-max plus a 5,000-run Monte Carlo.
- 5 Contagion & Lineage:** build the R0 blast-radius graph and Ingress → Nexus → Egress dataflow lineage.
- 6 Pareto Optimisation:** fill the developer-week budget with the highest risk-per-cost fixes; draw the efficient frontier.
- 7 Attest & Ship:** Merkle root, Ed25519 DSSE, negative certificate, then CycloneDX, SARIF, CISO report and CI gate.

THE MATH CORE: every score is a published formula, not a black box

MOSCA RUNWAY

$$Y_{\max} = (Z_{\text{reg}} - \text{now}) - X_{\text{eff}}$$

QUANTUM RISK

$$R_Q = \text{Overdue} \times P_{\text{HNDL}} \times (1 - \text{Agility}) \times W_{\text{intent}}$$

BLAST RADIUS

$$R_0(v) = |\text{Descendants}(G, v)|$$

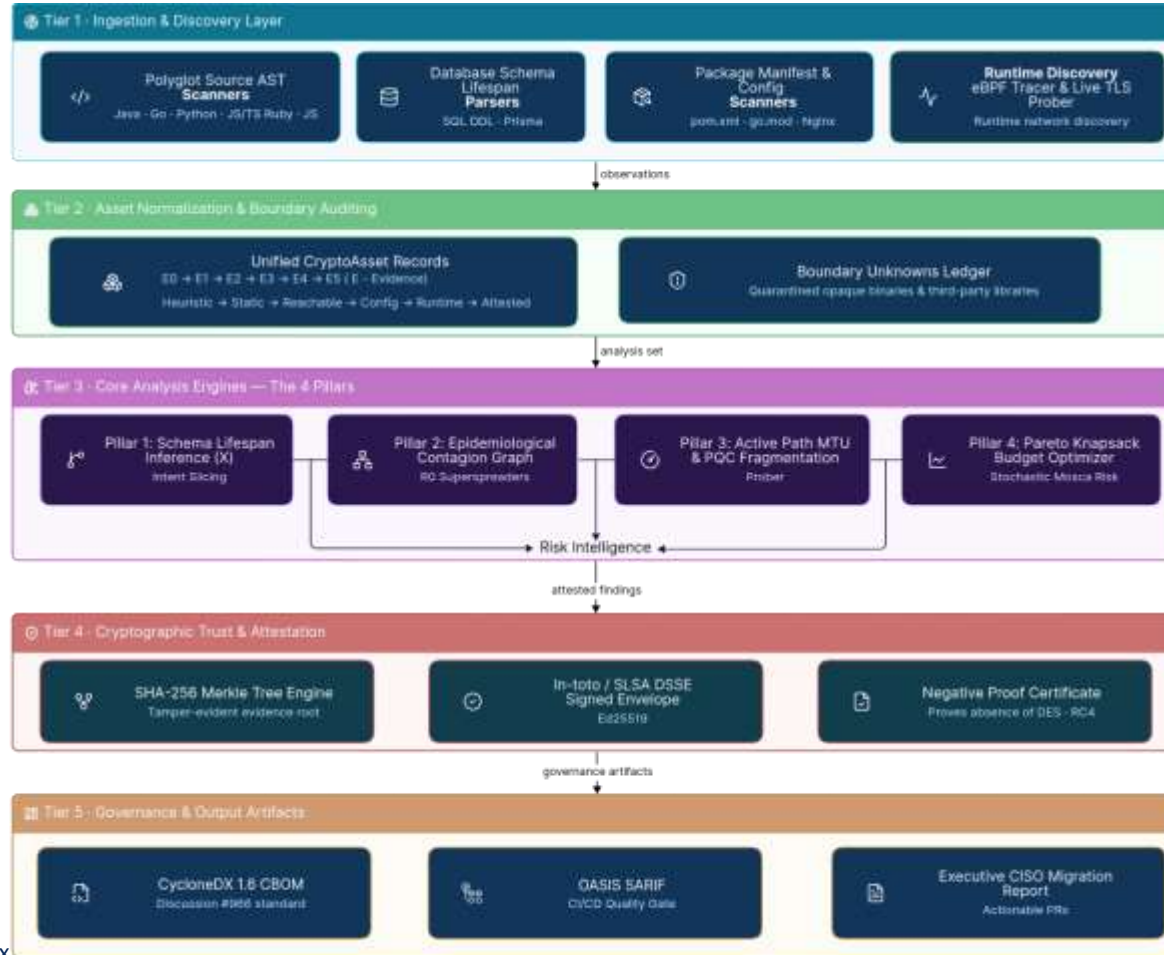
BREACH PROBABILITY

$$P = \text{mean}[1(\text{now} + X + Y > Z)] \times P_{\text{HNDL}}, N = 5,000$$

FIX EFFICIENCY

$$\eta = \Delta R / \text{Cost}, \Delta R = R_0 \times P_{\text{HNDL}} \times W_{\text{intent}} \times \text{RiskWeight}$$

System Architecture: 5 Tiers

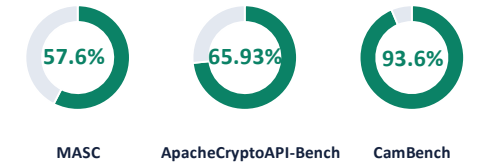


Technology Stack



Pydantic - NetworkX - Pygments - Starlette - D3.js - KaTeX - WebCrypto

Proof of Work: 3 Public Benchmarks



2,810 files - 762 cases - 89/89 regression tests



 Technical Feasibility

Already Built: 10-stage pipeline and 9 CLI commands (scan, gate, remediate, undo, probe, verify).

Benchmarked: 93.6-100% on 3 public suites; 89/89 regression tests pass.

Standards-Native: FIPS 203/204/205, CycloneDX, SARIF 2.1.0, in-toto / DSSE.

 Operational Feasibility

CI-Native: ecdat gate returns exit code 0 / 1 with SARIF pull-request annotations.

Zero-Network Mode: single-file, air-gapped report.html for isolated enclaves.

Safe by Default: dry-run diff, SHA-256 checks, undo / redo journal, protected targets.

 Financial Feasibility

Near-Zero Licence Cost: open-source Python stack; runs on a laptop or existing CI runner.

Public Funding Fit: DST/NQM backs financial assistance to take indigenous solutions to TRL-9 pilots.

Lean Pilot Budget: Rs [X] for CI runner + hosting [fill in].

 Social & Market Viability

Policy Tailwind: DST/NQM wants vendor CBOMs from FY 2027-28 and lists automated crypto-discovery tools for vendor compliance.

Preparedness Gap: India BFSI leaders rate PQC readiness 2.4 / 5 (ISB Institute of Data Science).

Atmanirbhar Fit: preferential market access for indigenous quantum-safe tools.

Challenge / Risk

Static analysis cannot be perfect (Rice's theorem); dynamic reflection hides algorithms.

Precision falls on messy production code (65.9% on the Apache corpus).

Auto-patching can break ciphertext compatibility.

Quantum arrival year is uncertain.

Opaque binaries, HSMs and third-party libraries.

Enterprise trust and adoption friction.

Our Strategy

→ E0-E5 evidence grading + Unknowns Ledger quarantine; tuned recall-first so nothing is silently dropped.

→ Dead-branch pruning and inter-procedural tracing; ambiguous flows go to a human-review tier.

→ Dry-run diff, SHA-256 pre / post check, atomic write, undo / redo journal, protected-target fence.

→ 5,000-run Monte Carlo on GRI-calibrated distributions; report P(breach) and VaR-95, not one date.

→ Boundary ledger + certificate / keystore scan + live TLS prober; certificates are scoped, never "absolute".

→ Open-standard outputs, CI-native flow, offline dashboard; no source code leaves the network.



Rollout Aligned to India's DST / NQM Milestones

1 FOUNDATIONS

CII 2027 | Enterprises 2028

Discover & inventory: signed CycloneDX CBOM per repository plus an Unknowns Ledger; feeds asset-repository and vendor-CBOM asks.

2 HIGH-PRIORITY MIGRATION

CII 2028 | Enterprises 2030

Prioritise & migrate: budget-fit sprint plan, MTU-safe ML-KEM / ML-DSA targets and 1-click fixes for priority systems.

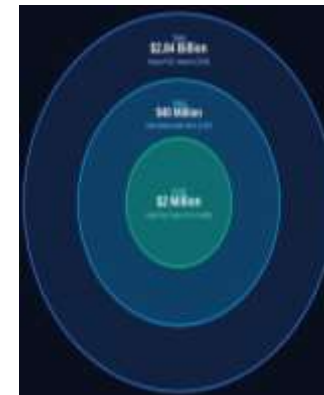
3 FULL PQC ADOPTION

CII 2029 | Enterprises 2033

Prove & monitor: negative certificates, fleet scorecard and a CI gate that blocks new classical-only deployments.



Market Sizing (sourced anchors)



TAM ~ USD 2.84 B (2030): global post-quantum cryptography market, MarketsandMarkets Oct 2025 (46.2% CAGR from USD 0.42 B in 2025); Mordor Intelligence puts it at USD 4.60 B.

SAM ~ USD 40 M (2030): India slice = TAM x 1.4%, India's share of world security spend (Gartner 2026: USD 3.4 B of USD 244.2 B). Derived estimate.

SOM ~ USD 2 M (2030): 5% of SAM. Our assumption; replace with the pilot pipeline.

TAM includes hardware, VPN and libraries; ECDAT serves the discovery and migration-planning slice, which analysts do not size separately.



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- 7 OWASP CycloneDX (ECMA-424): CBOM and Attestations (CDXA); community discussion #966 on PQC compliance.
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- 12 Afrose et al.: Evaluation of Static Vulnerability Detection Tools with Java Cryptographic API Benchmarks (ApacheCryptoAPI-Bench), IEEE TSE 2023.
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 Try It / See ItGITHUB : [ECDAT](#)